Week - 2 Module

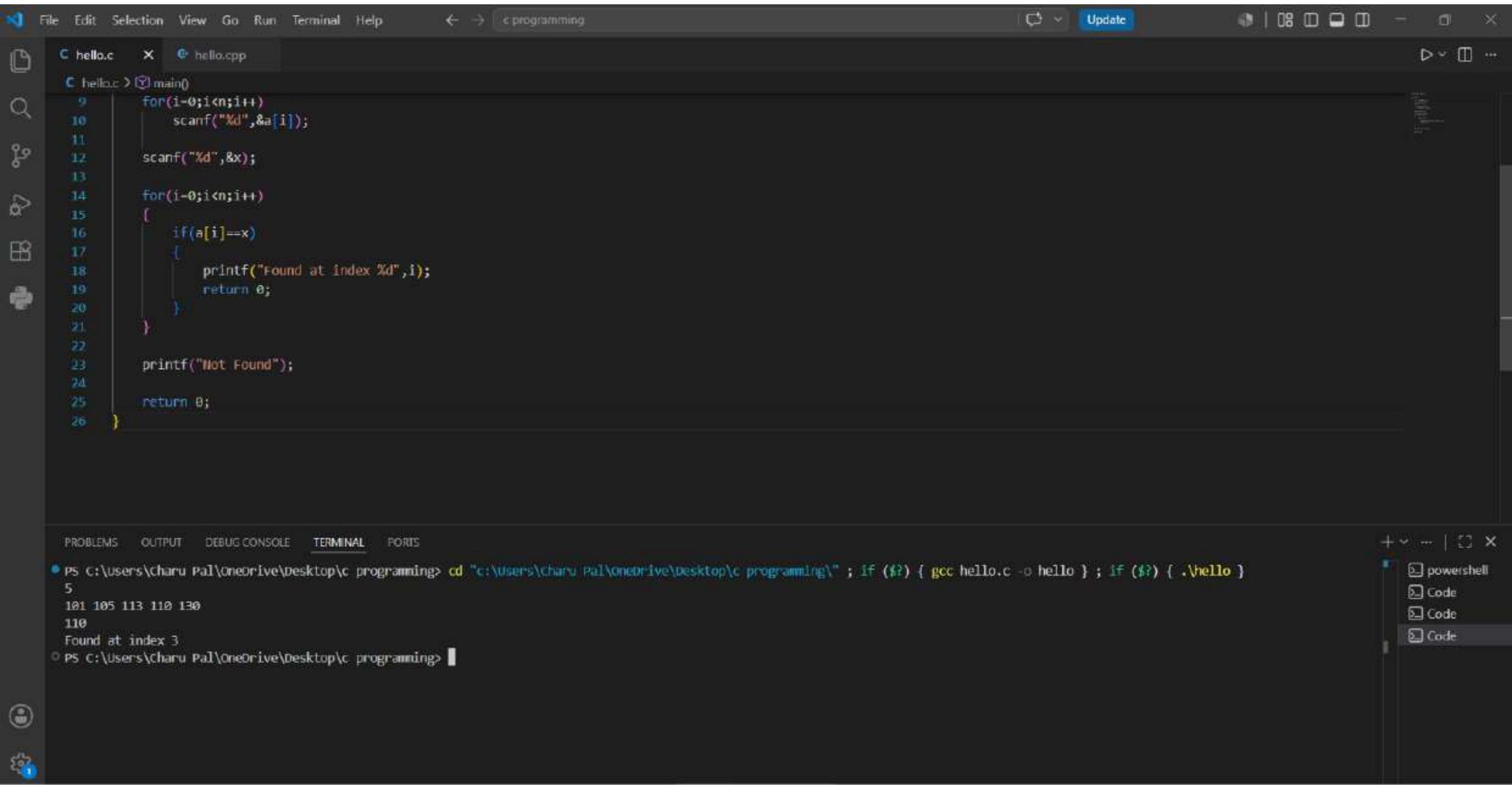
Ques : 1 find Student Roll no. (Linear Search).

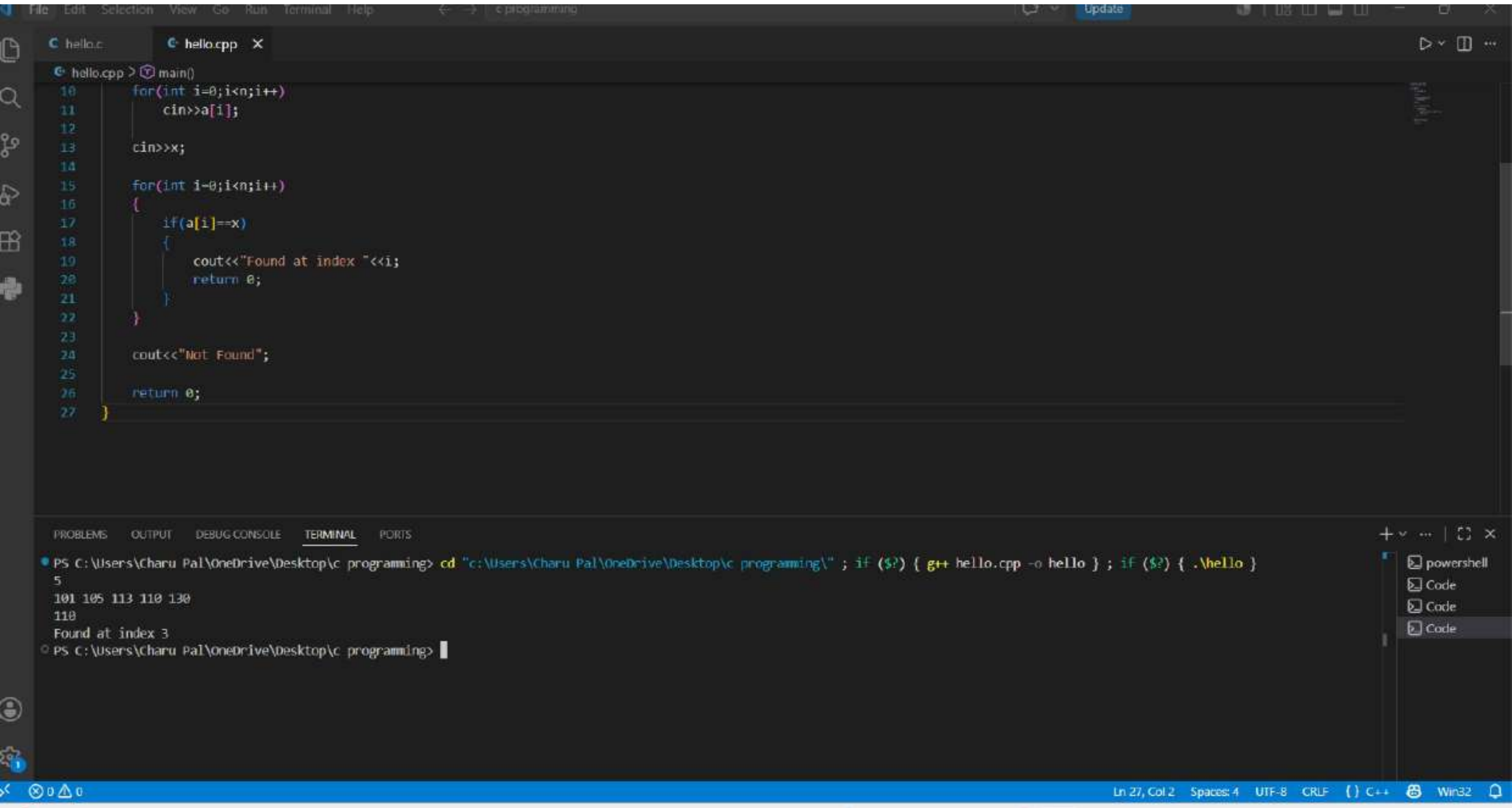
- C Program.

```c
# include <stdio.h>
int main(){
    int n, a[100], x, i;
    scanf("%d", &n);
    for(i=0; i<n; i++)
        scanf("%d", &a[i]);
    scanf("%d", &x);
    for(i=0; i<n; i++){
        if(a[i]==x){
            printf("found at index %d", i);
            return 0;
        }
    }
    printf("Not found");
    return 0;
}.
```

- C++ Program

```cpp
# include <iostream>
using namespace std;
int main(){
    int n, a[100], x;
    cin >> n;
    for(int i=0; i<n; i++)
        cin >> a[i];
    cin >> x;
    for(int i=0; i<n; i++){
        if(a[i]==x){
            cout << "found at index" << i;
            return 0;
        }
    }
    cout << "Not found";
    return 0;
}
```


File Edit Selection View Go Run Terminal Help
c programming
Update
hello.c
hello.cpp
hello.c > main()
for(i=0;i<n;i++)
scanf("%d",&a[i]);
scanf("%d",&x);
for(i=0;i<n;i++)
{
if(a[i]==x)
{
printf("Found at index %d",i);
return 0;
}
}
printf("Not Found");
return 0;
}
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
PS C:\Users\Charu Pal\OneDrive\Desktop\c programming> cd "c:\Users\Charu Pal\OneDrive\Desktop\c programming\" ; if ($?) { gcc hello.c -o hello } ; if ($?) { .\hello }
5
101 105 113 110 130
110
Found at index 3
PS C:\Users\Charu Pal\OneDrive\Desktop\c programming>
powershell
Code
Code
Code


File Edit Selection View Go Run Terminal Help
Update
hello.c
hello.cpp
hello.cpp > main()
for(int i=0;i<n;i++)
cin>>a[i];
cin>>x;
for(int i=0;i<n;i++)
{
if(a[i]==x)
{
cout<<"Found at index "<<i;
return 0;
}
}
cout<<"Not Found";
return 0;
}
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
PS C:\Users\Charu Pal\OneDrive\Desktop\c programming> cd "c:\Users\Charu Pal\OneDrive\Desktop\c programming\" ; if ($?) { g++ hello.cpp -o hello } ; if ($?) { .\hello }
5
101 105 113 110 130
110
Found at index 3
PS C:\Users\Charu Pal\OneDrive\Desktop\c programming>
powershell
Code
Code
Code
Ln 27, Col 2 Spaces: 4 UTF-8 CRLF {} C++ Win32

Date:
Page:

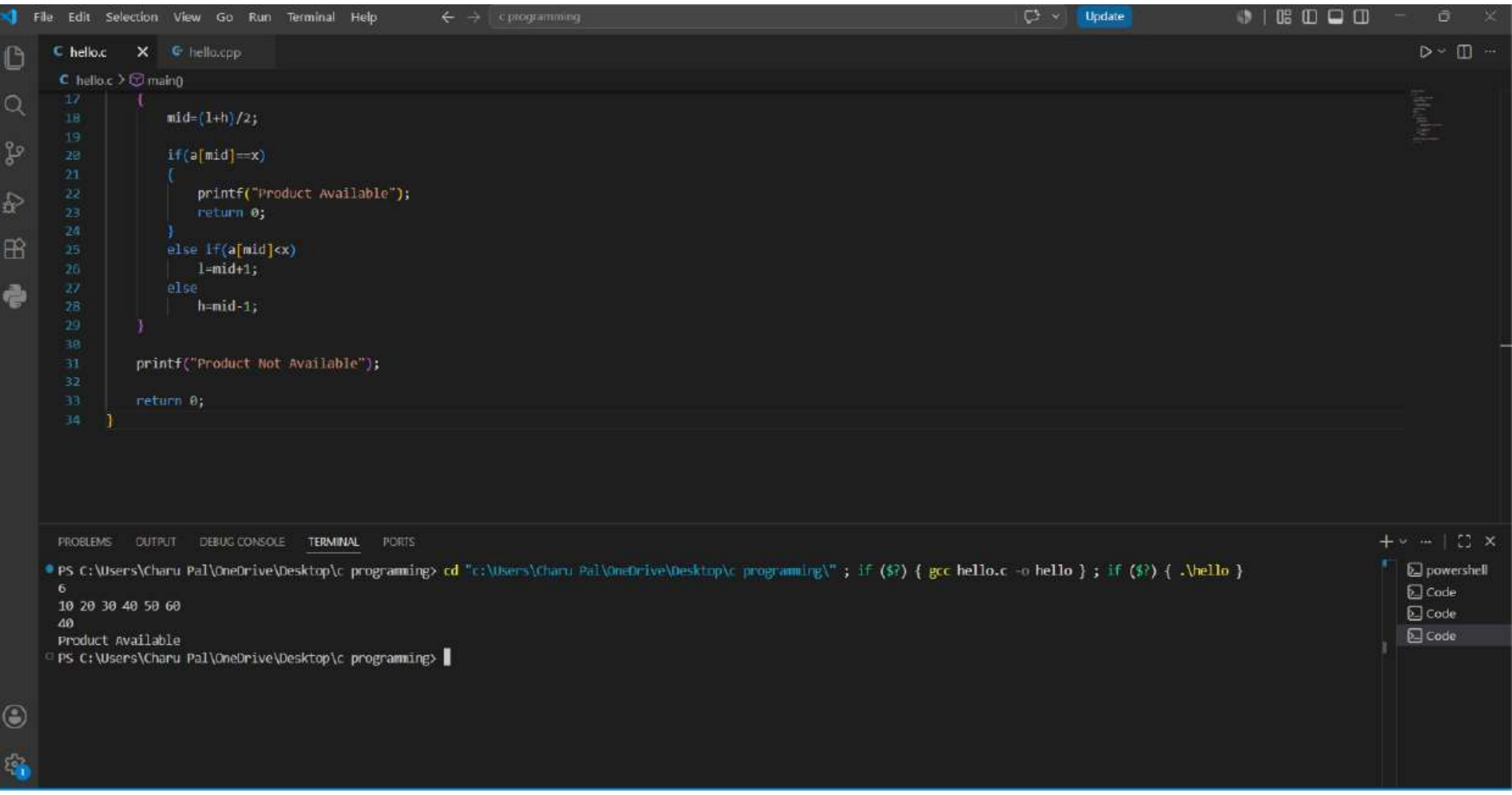
Ques: 2 Search Product ID (Binary Search)

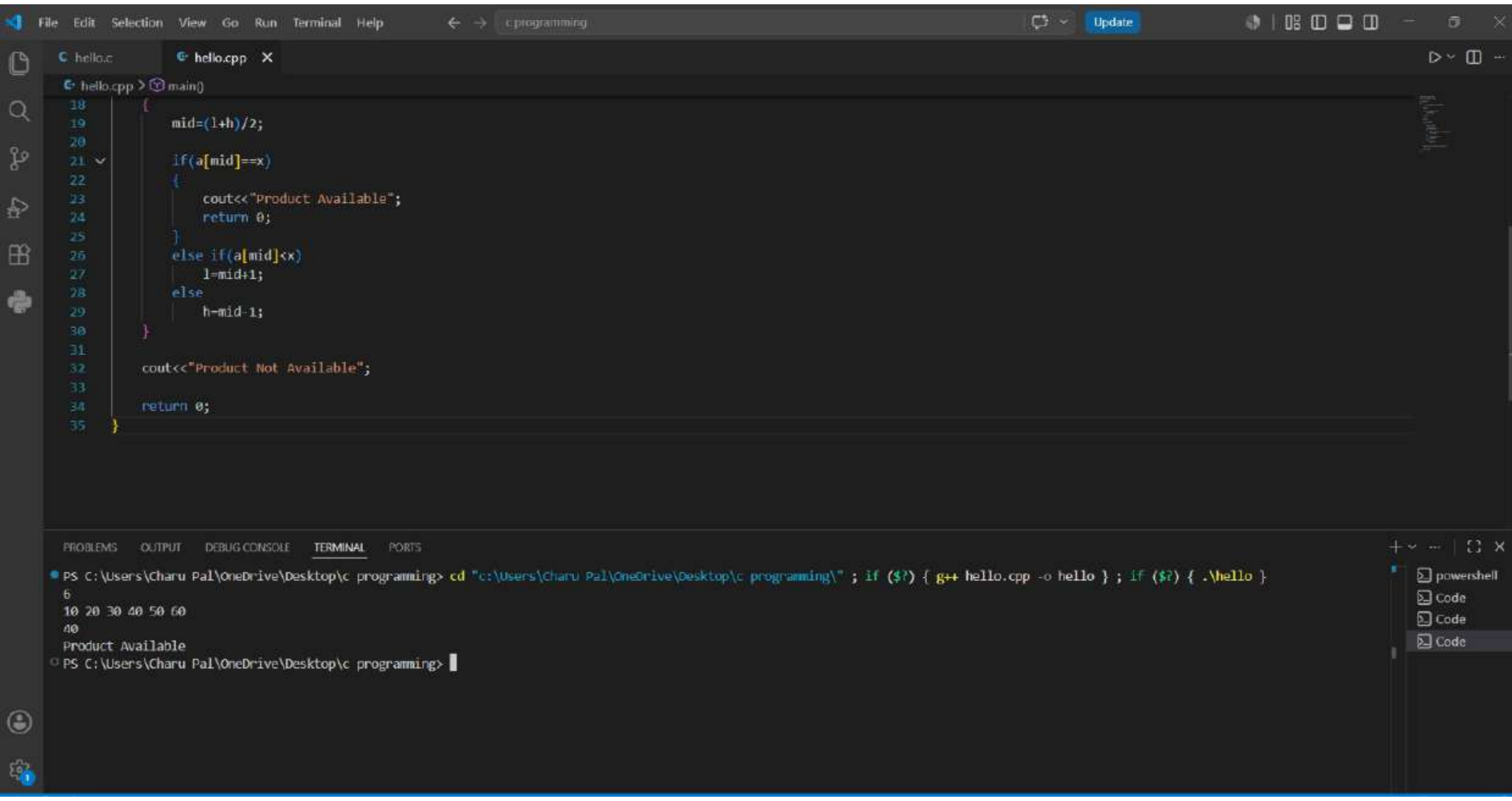
C Program

```c
# include <stdio.h>
int main(){
    int n, a[100], x, l=0, h, mid;
    scanf("%d", &n);
    for (int i=0; i<n; i++)
        scanf ("%d", &a[i]);
    scanf ("%d", &x);
    h = n-1;
    while (l<=h){
        mid = (l+h) /2;
        if(a[mid] == x) {
            printf("Product Available");
            return 0;
        }
        else if (a[mid] < x)
            l = mid+1;
        else
            h = mid - 1;
    }
    printf ("Product Not Available");
    return 0;
}
```

C++ Program.

```cpp
# include <iostream>
using namespace std;
int main () {
    int n, a[100], x, l=0, h, mid;
    cin >> n;
    for (int i=0; i<n; i++)
        cin >> a[i];
    cin >> x;
    h=n-1;
    while (l<=h) {
        mid =(l+h)/2;
        if (a[mid] == x) {
            cout << "Product Available";
            return 0;
        }
        else if (a[mid < x)
            l = mid +1;
        else
            h = mid - 1;
    }
    cout << "Product Not Available";
    return 0;
}
```


File Edit Selection View Go Run Terminal Help
c programming
Update
hello.c
hello.cpp
hello.c > main()
{
mid=(l+h)/2;
if(a[mid]==x)
{
printf("Product Available");
return 0;
}
else if(a[mid]<x)
l=mid+1;
else
h=mid-1;
}
printf("Product Not Available");
return 0;
}
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
PS C:\Users\Charu Pal\OneDrive\Desktop\c programming> cd "c:\Users\Charu Pal\OneDrive\Desktop\c programming\" ; if ($?) { gcc hello.c -o hello } ; if ($?) { .\hello }
6
10 20 30 40 50 60
40
Product Available
PS C:\Users\Charu Pal\OneDrive\Desktop\c programming>
powershell
Code
Code
Code


File Edit Selection View Go Run Terminal Help
c programming
Update
hello.c
hello.cpp
hello.cpp > main()
{
mid=(l+h)/2;
if(a[mid]==x)
{
cout<<"Product Available";
return 0;
}
else if(a[mid]<x)
l=mid+1;
else
h=mid-1;
}
cout<<"Product Not Available";
return 0;
}
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
PS C:\Users\Charu Pal\OneDrive\Desktop\c programming> cd "c:\Users\Charu Pal\OneDrive\Desktop\c programming\" ; if ($?) { g++ hello.cpp -o hello } ; if ($?) { .\hello }
6
10 20 30 40 50 60
40
Product Available
PS C:\Users\Charu Pal\OneDrive\Desktop\c programming>
powershell
Code
Code
Code

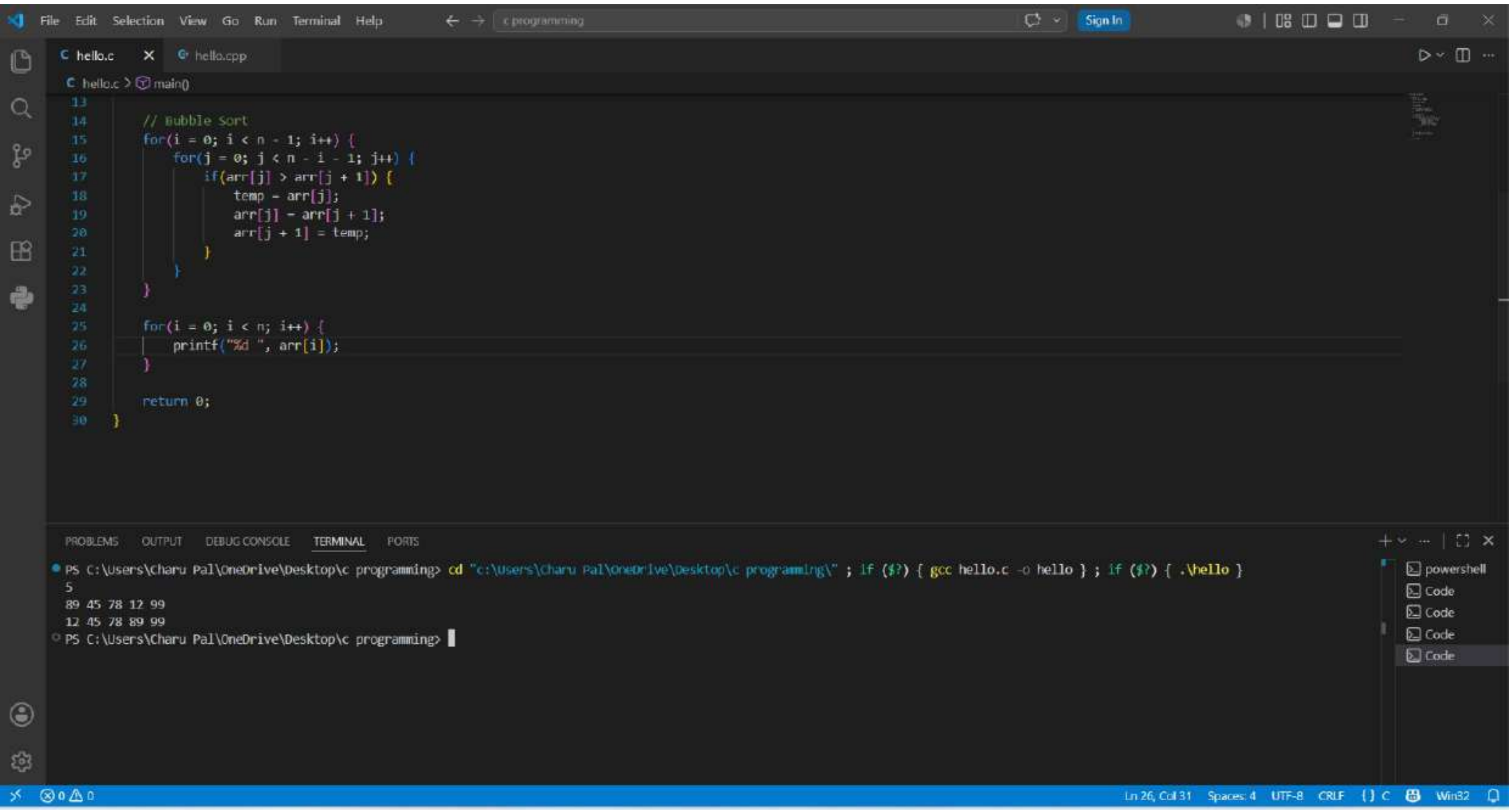
Ques: 3 Arrange Exam Scores (Bubble Sort)

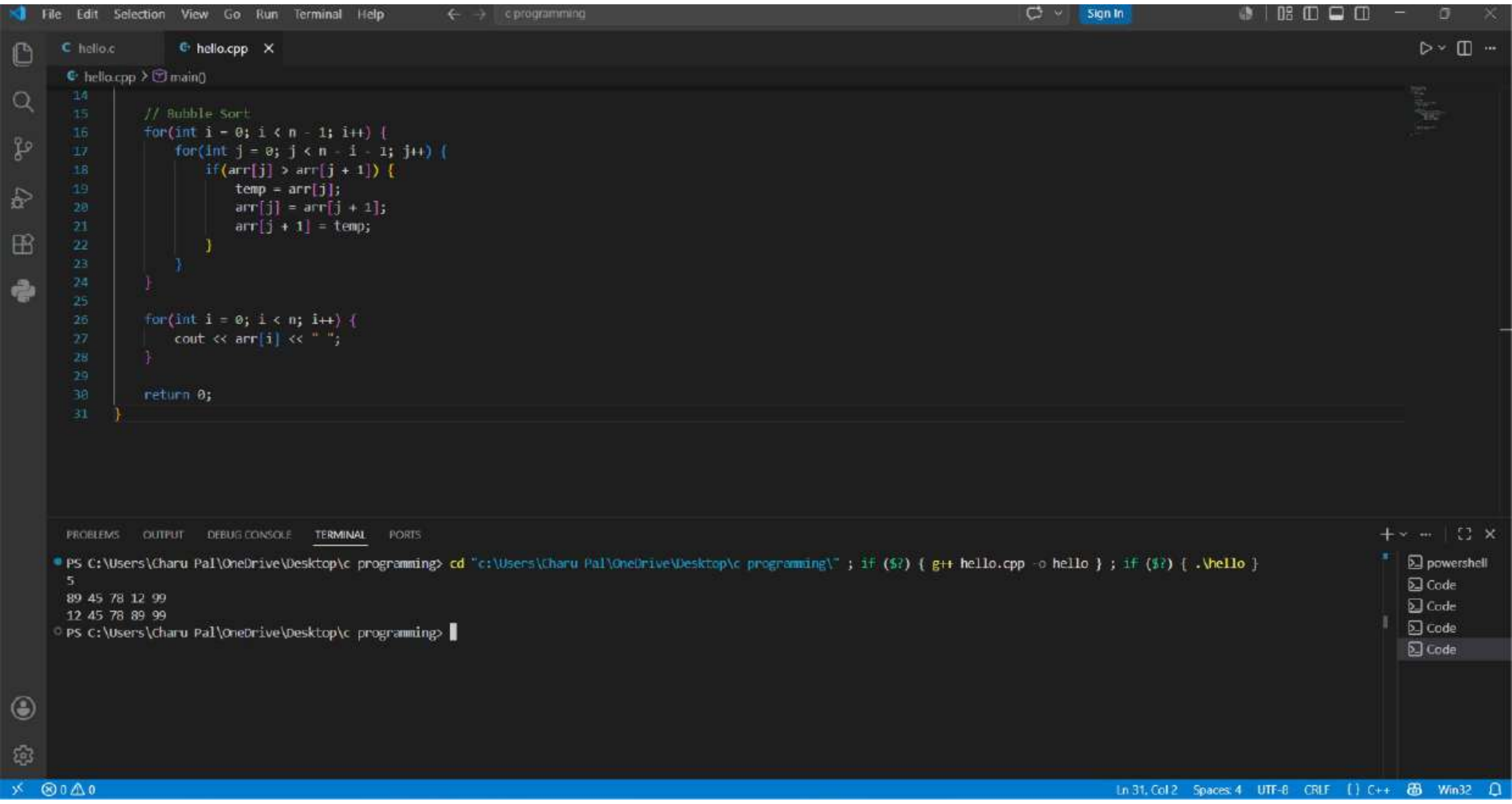
- C Program.

```c
# include <stdio.h>
int main(){
    int n, i, j, temp;
    scanf("%d", &n);
    int arr[n];
    for (i=0; i<n; i++) {
        for(j=0; j<n-i-1; j++){
            if( arr[j] > arr[j+1]){
                temp = arr[j];
                arr[j] = arr[j+1];
                arr[j+1] = temp;
            }
        }
    }
    for (i=0; i<n; i++){
        printf("%d", arr[i]);
    }
    return 0;
}
```

- C++ Program

```cpp
# include <iostream>
using namespace std;
int main(){
    int n, temp;
    cin >> n;
    int arr[n];
    for(int i=0; i<n; i++){
        cin >> arr[i];
    }
    // Bubble Sort
    for(int i=0; i<n-1; i++){
        for (int j=0; j<n-i-1; j++){
            if (arr[j] > arr[j+1]){
                temp = arr[j];
                arr[j] = arr[j+1];
                arr[j+1] = temp;
            }
        }
    }
    for (int i=0; i<n; i++){
        cout << arr[i] << " ";
    }
    return 0;
}
```


File Edit Selection View Go Run Terminal Help
c programming
Sign In
hello.c
hello.cpp
hello.c > main()
// Bubble Sort
for(i = 0; i < n - 1; i++) {
for(j = 0; j < n - i - 1; j++) {
if(arr[j] > arr[j + 1]) {
temp = arr[j];
arr[j] = arr[j + 1];
arr[j + 1] = temp;
}
}
}
for(i = 0; i < n; i++) {
printf("%d ", arr[i]);
}
return 0;
}
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
PS C:\Users\Charu Pal\OneDrive\Desktop\c programming> cd "c:\Users\Charu Pal\OneDrive\Desktop\c programming\" ; if ($?) { gcc hello.c -o hello } ; if ($?) { .\hello }
5
89 45 78 12 99
12 45 78 89 99
PS C:\Users\Charu Pal\OneDrive\Desktop\c programming>
powershell
Code
Code
Code
Code
Ln 26, Col 31 Spaces: 4 UTF-8 CRLF {} C Win32



File Edit Selection View Go Run Terminal Help
c programming
Sign In
hello.c
hello.cpp
hello.cpp > main()
// Bubble Sort
for(int i = 0; i < n - 1; i++) {
for(int j = 0; j < n - i - 1; j++) {
if(arr[j] > arr[j + 1]) {
temp = arr[j];
arr[j] = arr[j + 1];
arr[j + 1] = temp;
}
}
}
for(int i = 0; i < n; i++) {
cout << arr[i] << " ";
}
return 0;
}
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
PS C:\Users\Charu Pal\OneDrive\Desktop\c programming> cd "c:\Users\Charu Pal\OneDrive\Desktop\c programming\" ; if ($?) { g++ hello.cpp -o hello } ; if ($?) { .\hello }
5
89 45 78 12 99
12 45 78 89 99
PS C:\Users\Charu Pal\OneDrive\Desktop\c programming>
powershell
Code
Code
Code
Code
Ln 31, Col 2 Spaces: 4 UTF-8 CRLF {} C++ Win32

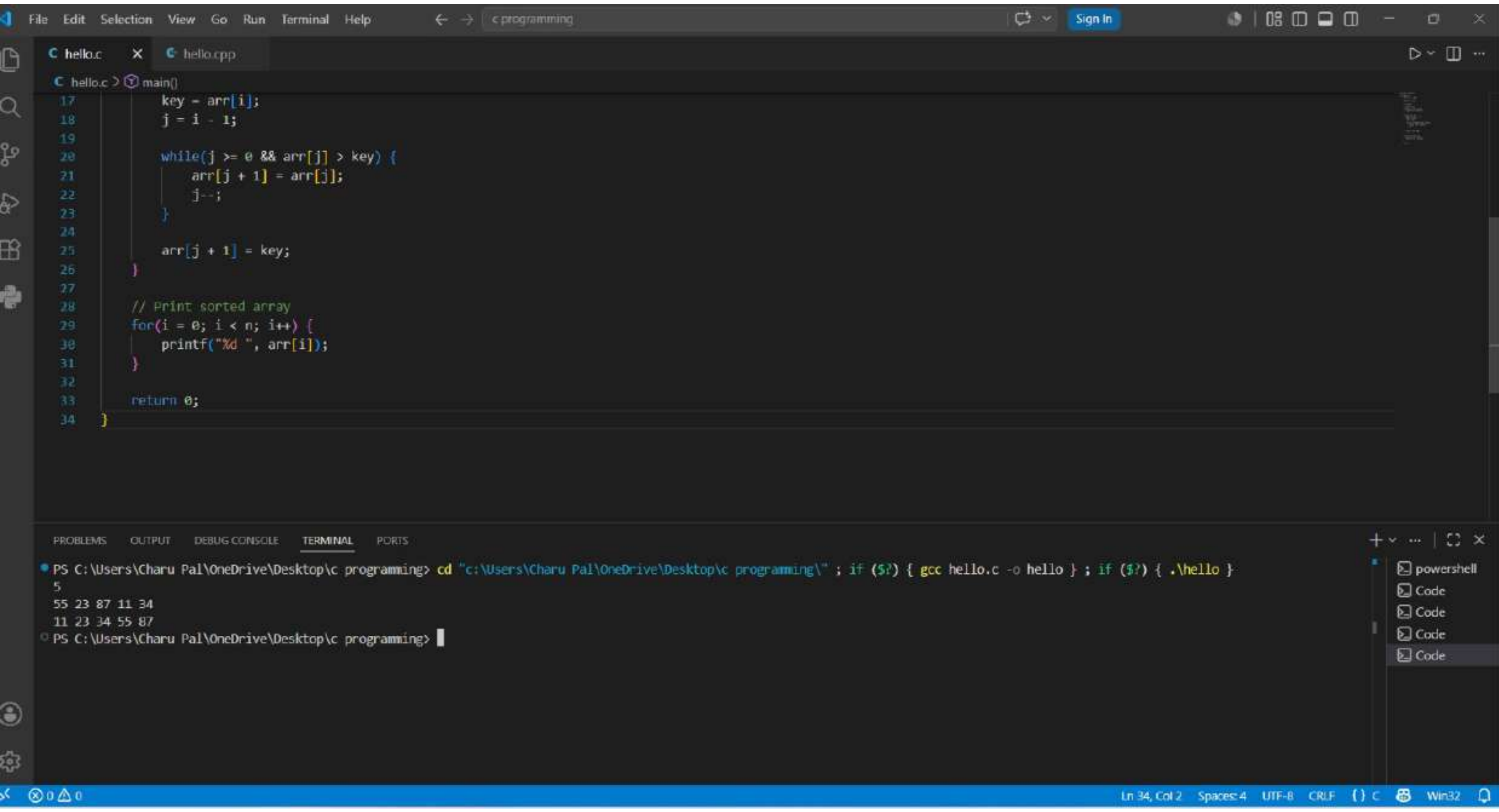
Ques: Arrange library book Numbers (Insertion Sort)

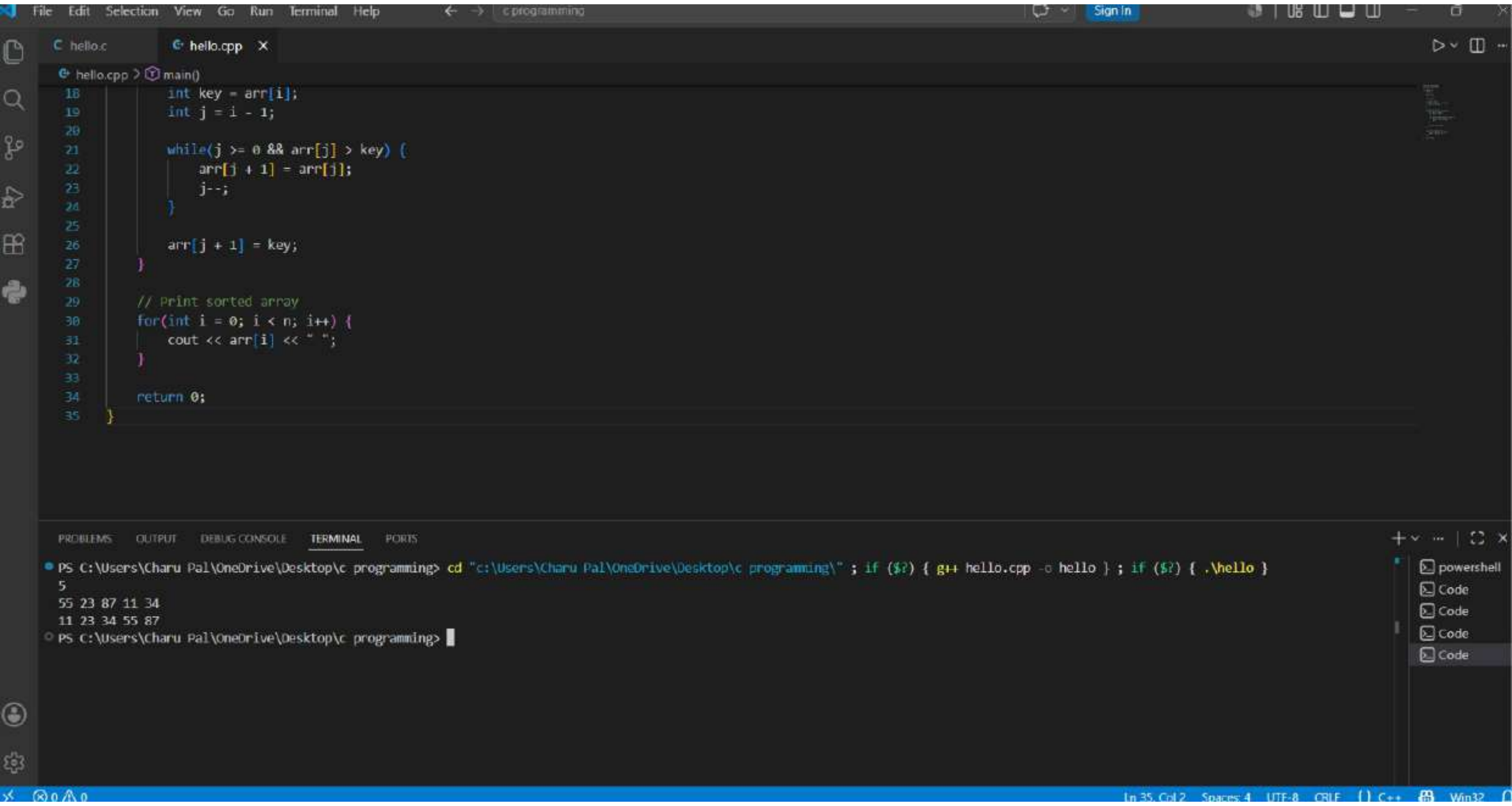
- C Program.

```c
# include <stdio.h>
int main(){
    int n, i, j, key;
    scanf("%d",&n);
    int arr[n];
    // Input array
    for(i=0; i<n; i++){
        key=arr[i];
        j = i-1;
        while(j>=0 &&arr[j]>key)
        {
            arr[j+1]=arr[j];
            j--;
        }
        arr[j+1]= key;
    }
    // Print Sorted array
    for (i=0; i<n; i++){
        printf("%d", arr[i]);
    }
    return 0;
}
```

- C++ Program.

```cpp
# include <iostream>
using namespace std;
int main(){
    int n;
    cin >> n;
    // input array
    for(int i=0; i<n ; i++){
        cin >> arr[i];
    }
    // Insertion sort
    for(int i=1; i<n ; i++){
        int key = arr[i];
        int j= i-1;
        while (j>=0 && arr[j]>key){
            arr[j+1] = arr[j];
            j--;
        }
        arr [j+1] = key;
    }
    // Print Sorted array
    for (int i=0; i<n; i++){
        cout << arr[i] <<" ";
    }
    return 0;
}
```


File Edit Selection View Go Run Terminal Help
c programming
Sign In
hello.c
hello.cpp
hello.c > main()
key = arr[i];
j = i - 1;
while(j >= 0 && arr[j] > key) {
arr[j + 1] = arr[j];
j--;
}
arr[j + 1] = key;
}
// Print sorted array
for(i = 0; i < n; i++) {
printf("%d ", arr[i]);
}
return 0;
}
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
PS C:\Users\Charu Pal\OneDrive\Desktop\c programming> cd "c:\Users\Charu Pal\OneDrive\Desktop\c programming\" ; if ($?) { gcc hello.c -o hello } ; if ($?) { .\hello }
5
55 23 87 11 34
11 23 34 55 87
PS C:\Users\Charu Pal\OneDrive\Desktop\c programming>
powershell
Code
Code
Code
Code


File Edit Selection View Go Run Terminal Help
c programming
Sign In
hello.c
hello.cpp
hello.cpp > main()
int key = arr[i];
int j = i - 1;
while(j >= 0 && arr[j] > key) {
arr[j + 1] = arr[j];
j--;
}
arr[j + 1] = key;
}
// Print sorted array
for(int i = 0; i < n; i++) {
cout << arr[i] << " ";
}
return 0;
}
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
PS C:\Users\Charu Pal\OneDrive\Desktop\c programming> cd "c:\Users\Charu Pal\OneDrive\Desktop\c programming\" ; if ($?) { g++ hello.cpp -o hello } ; if ($?) { .\hello }
5
55 23 87 11 34
11 23 34 55 87
PS C:\Users\Charu Pal\OneDrive\Desktop\c programming>
powershell
Code
Code
Code
Code
Ln 35, Col 2 Spaces: 4 UTF-8 CRLF {} C++ Win32

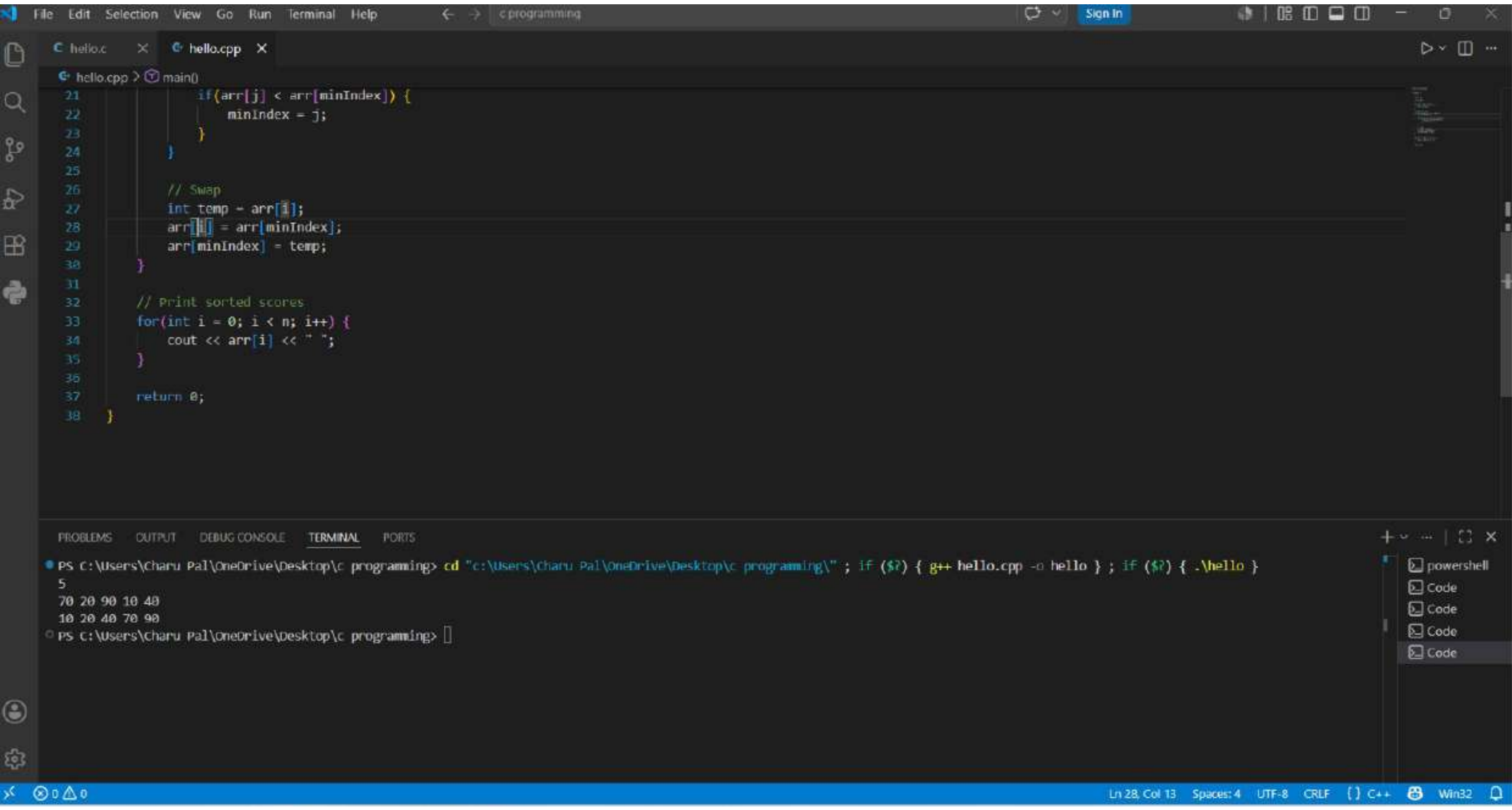
Ques 5. Rank Players by Score (Selection Sort)

- C Program

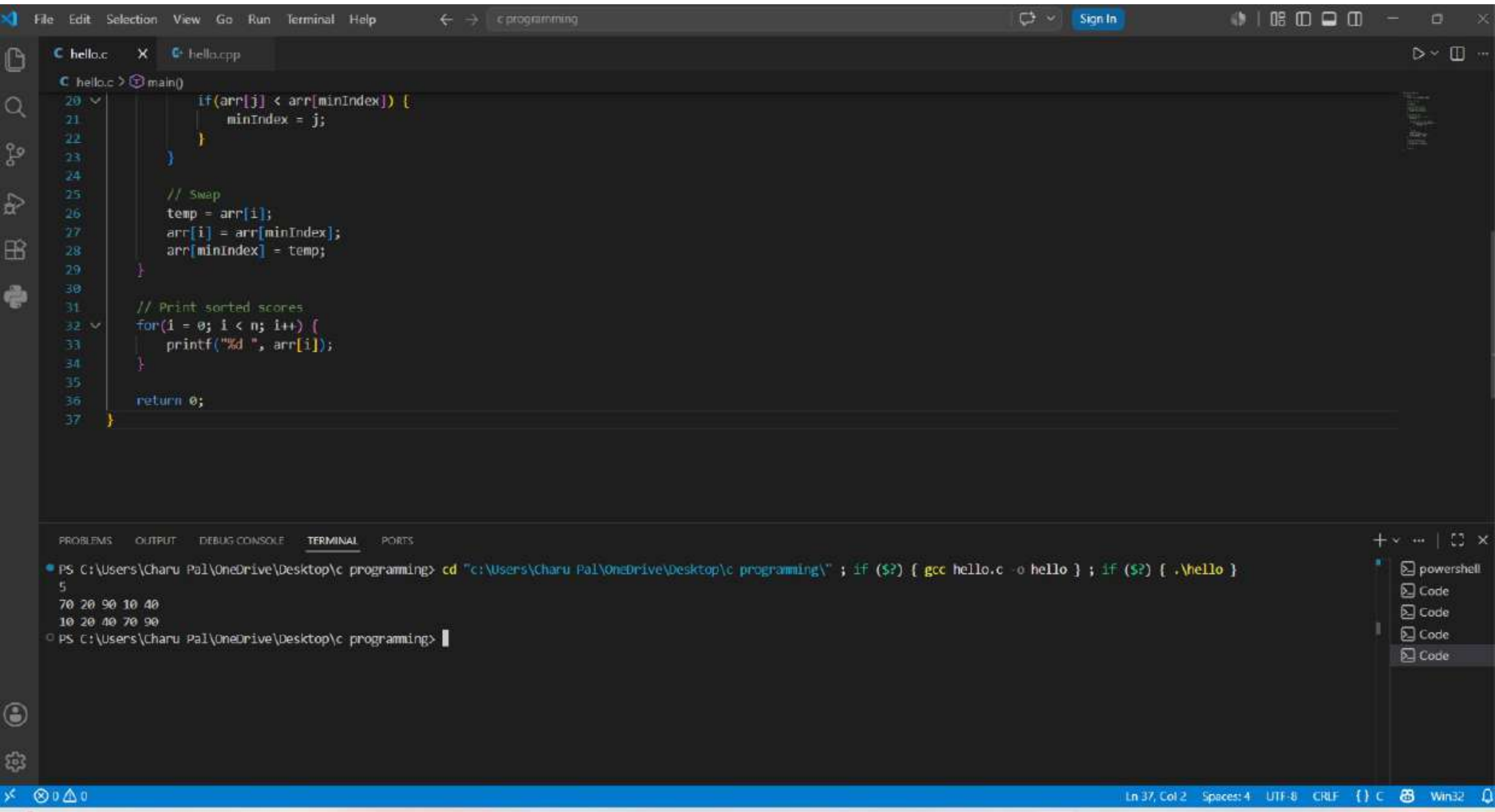
```c
#include<stdio.h>
int main( ){
    int n, i, j, minIndex, temp;
    scanf("%d", &n);
    int arr[n];
    // Input player score.
    for(i=0; i<n; i++){
        scanf("%d", &arr[i]);
    }
    // Selection sort
    for(i=0; i<n-1; i++){
        minIndex = i;
        for(j=i+1; j<n; j++){
            if(arr[j]<arr[min]){
                minIndex = j;
            }
        }
        // Swap.
        temp = arr[i];
        arr[i] = arr[minIndex];
        arr[minIndex] = temp;
    }
    // print sorted scores
    for(i=0; i<n; i++){
        printf("%d", arr[i]);
    }
    return 0;
}.
```

- C++ Program.

```cpp
#include <iostream>
using namespace std;
int main(){
    int n;
    cin >> n;
    int arr[n];
    // Input player scores
    for(int i=0; i<n; i++){
        cin >> arr[i];
    }
    // Selection sort
    for(int i=0; i<n-1; i++){
        int minIndex = i;
        for(int j=i+1; j<n; j++){
            if(arr[j] < arr[min]){
                minIndex = j;
            }
        }
        // Swap.
        int temp = arr[i];
        arr[i] = arr[minIndex];
        arr[minIndex] = temp;
    }
    // print Sorted scores
    for(int i=0; i<n; i++){
        cout << arr[i] << " ";
    }
    return 0;
}
```



File Edit Selection View Go Run Terminal Help
c programming
Sign In
hello.c
hello.cpp
hello.cpp > main()
if(arr[j] < arr[minIndex]) {
minIndex = j;
}
}
// Swap
int temp = arr[i];
arr[i] = arr[minIndex];
arr[minIndex] = temp;
}
// Print sorted scores
for(int i = 0; i < n; i++) {
cout << arr[i] << " ";
}
return 0;
}
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
PS C:\Users\Charu Pal\OneDrive\Desktop\c programming> cd "c:\Users\Charu Pal\OneDrive\Desktop\c programming\" ; if ($?) { g++ hello.cpp -o hello } ; if ($?) { .\hello }
5
70 20 90 10 40
10 20 40 70 90
PS C:\Users\Charu Pal\OneDrive\Desktop\c programming>
powershell
Code
Code
Code
Code
Ln 28, Col 13 Spaces: 4 UTF-8 CRLF { } C++ Win32


File Edit Selection View Go Run Terminal Help
c programming
Sign In
hello.c
hello.cpp
hello.c > main()
if(arr[j] < arr[minIndex]) {
minIndex = j;
}
}
// Swap
temp = arr[i];
arr[i] = arr[minIndex];
arr[minIndex] = temp;
}
// Print sorted scores
for(i = 0; i < n; i++) {
printf("%d ", arr[i]);
}
return 0;
}
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
PS C:\Users\Charu Pal\OneDrive\Desktop\c programming> cd "c:\Users\Charu Pal\OneDrive\Desktop\c programming\" ; if ($?) { gcc hello.c -o hello } ; if ($?) { .\hello }
5
70 20 90 10 40
10 20 40 70 90
PS C:\Users\Charu Pal\OneDrive\Desktop\c programming>
powershell
Code
Code
Code
Code
Ln 37, Col 2 Spaces: 4 UTF-8 CRLF { } C Win32